

YAUHENI SAMCHUK

+375(33) 320-7004 ◊ Minsk, Belarus

[Kaggle](#) ◊ [Linkedin](#) ◊ [GitHub](#)

OBJECTIVE

AI / Machine Learning Engineer with **3+ years of experience** building and deploying production AI systems across **LLM/RAG, NLP, computer vision, speech processing and classical ML**. Experienced in developing **end-to-end AI solutions** spanning model development, prompt engineering, backend integration, evaluation, deployment, and monitoring. Strong **Python** background with hands-on experience in **LLM APIs, RAG, Docker, FastAPI, message-driven architectures and on-premise AI deployment**.

EDUCATION

Bachelor of Computer Science, Belarussian State University of Informatics and Radioelectronics

Artificial Intelligence

2020- 2024

EXPERIENCE

Machine Learning Engineer, Taqtile

- **Warehouse Inventory Management (CV)**: Developed a real-time object detection and counting system by fine-tuning a **YOLOv8** model on a custom augmented dataset. Achieved strong production-grade performance with **90% Precision** and **80% Recall** on a held-out test set. Optimized the model via **ONNX Runtime** with INT8 quantization, reducing inference **latency by 30%** with negligible accuracy drop, enabling seamless real-time deployment directly on **on-premise edge** hardware.
- Developed and deployed a **real-time employee activity monitoring system** using **YOLOX** for person and phone detection and **DeepSORT** for multi-object tracking. Designed a **spatio-temporal interaction detection pipeline** to associate phones with tracked employees and identify phone usage during production. Optimized inference with **ONNX** and deployed the solution on an **on-premise edge** device within the customer's local infrastructure.
- Fine-tuned an **LLaMA-7B** model with **QLoRA** on a proprietary customer QA dataset, utilizing a local NVIDIA RTX 3090 GPU. Achieved **80% exact-match accuracy** and **ROUGE-L score of 0.8** on a held-out test set. Deployed the model as a **FastAPI** microservice and exposed this service externally via **ngrok** tunnel for secure remote access and testing.
- Developed a **client-side browser extension** that performs real-time speech-to-text transcription using **Whisper** model, eliminating the need for external cloud APIs. To enable **client-side inference**, quantized the Whisper model to INT8 precision and deployed it via **ONNX Runtime Web**. Built with **JavaScript, Webpack, and ONNX Runtime**.

March, 2023 - February, 2024

Machine Learning Engineer, Belgazprombank

- Developed and deployed **customer churn prediction models** for individuals and legal entities across **transactions, deposits, and loans**. Compared **gradient boosting, Transformers, RNNs, and other classification approaches**, with evaluation focused on **Precision, Recall, and PR-AUC** due to the highly imbalanced churn target. Built an end-to-end **Prefect** pipeline covering data extraction, preprocessing, model inference, post-processing, and delivery of predictions to **Power BI**, with separate validation flows for **continuous model-quality monitoring**. Integrated model predictions into the bank's **customer retention workflow**, prioritizing high-risk customers for outbound calls based on **predicted churn probability and customer value (LTV-like metric)**. Designed an **A/B testing framework** comparing the existing customer-selection process with **ML-driven targeting**, evaluating the impact using **retention rate and incremental retained revenue**.

- Developed an end-to-end **speech analytics pipeline** to automate previously manual **call-quality control** and analysis. Built the complete processing chain from **audio preprocessing with FFmpeg, VAD, and noise reduction** through local **Whisper ASR** and speaker diarization with **pyannote-audio** to a local **LLM** for automated call summarization, extraction of contact reasons, and quality assessment (greeting, farewell, problem resolution). Designed the **LangChain**-based orchestration, including prompt engineering, structured outputs, and post-processing, and prepared the results for **Power BI** reporting. **Evaluated ASR with WER, diarization with DER, and LLM outputs using accuracy for structured fields and manual evaluation for open-ended results**. Deployed the solution on the **bank's on-premise server**; 2–4 minute recordings were processed in approximately **up to 10 seconds**.

March, 2024 - August, 2025

Python AI Engineer, STA Logistics

- Developed and deployed **15+ production AI services** integrating **OpenAI, Anthropic, and other LLM providers**, covering document processing, summarization, transcription, assistants, and business automation. The deployed solutions generated an estimated **400–500 hours and €5,000 in monthly savings** based on actual usage and department-specific labor costs.
- Worked closely with **business stakeholders and end users** to analyze existing workflows, identify automation opportunities, and translate business requirements into technical solutions. Managed **iterative requirement changes** throughout development, adapting system architecture and implementation to evolving business needs while maintaining **SOLID principles, modularity, and maintainability**.
- Designed and implemented the **backend and AI infrastructure** for LLM-powered services, including API integrations, request orchestration, retries, output validation, fallback strategies, and reliable task processing using **Python, RabbitMQ, MongoDB, and Docker**. Developed **LLM prompting and model-selection workflows** across GPT, Claude, and open-source models.
- Developed a **production RAG system** for an internal corporate messenger using **Qdrant** and a **local Russian-language embedding model**. Improved chunking and retrieval strategies, built the backend and messenger integration, and implemented evaluation with **RAGAS**, including **Context Precision** and **Context Recall**.

August, 2025 - up to the present time

SKILLS

- **Programming:** Python, SQL, C++, R
- **ML / AI:** PyTorch, TensorFlow, scikit-learn, CatBoost, Transformers, LLMs, RAG, Computer Vision, Object Detection, Object Tracking, NLP, ASR, VAD
- **LLM / Generative AI:** OpenAI API, Anthropic API, LangChain, Prompt Engineering, RAG, Structured Outputs, LLM Evaluation, RAGAS
- **Data / Backend:** Pandas, NumPy, FastAPI, Flask, RabbitMQ, MongoDB, Qdrant, MSSQL
- **MLOps / Deployment:** Docker, GitHub Actions, GitLab CI/CD, MLflow, ONNX, Model Quantization, Prefect, Linux

Languages

- English – intermediate.
- Polish – intermediate.